

Using eBird Data for Ecology and Conservation - Day 4 - Thursday, 25th September 2025

0. Here is a tentative timeline of activities for the workshop:

Time	Activity
Before the workshop	Share resources to participants to be prepared for the workshop, along with a trial dataset and an ecological question from the organizers. The participants are expected to come up with an approach to answer the question. They will also be asked to think about possible questions answerable through eBird data and send it to us.
0:00–0:30	A walkthrough through the best practices for using eBird data, focused on the biases of data collection and reasons for data cleaning.
0:30–1:00	Go through the questions from the participants and review the answerability through eBird data.
1:00–1:30	Select one participant question and guide the participants with the code for basic analysis and visualisation.
1:30–2:00	Interactively elaborate on the approach to answer our pre-shared question.

1. What Do the Participants Stand to Gain?

1. Learn to formulate meaningful ecological questions using eBird data.
2. limitations of the data and best practices to work with eBird data.
3. Gain hands-on experience with data downloading, cleaning, and visualization.

2. Prerequisite reading resources:

Resources for participants to be prepared for the workshop

- a. Introduction to R: <https://bit.ly/R-preparation>

- i. Understanding this document will help in cleaning the rusty engine (Participants should ideally have a laptop with R and Rstudio)
- b. **Analysing eBird data using R:** Watch this videos to get a walkthrough of downloading eBird data, and running simple cleaning and analysis steps.
 - i. Part 1 - <https://www.youtube.com/watch?v=jBGVy7K7dH8> (Required)
 - ii. Part 2 - <https://www.youtube.com/watch?v=fYldtekOrZI> (Optional)

Once you have watched these, your car is ready to be airdropped and hit the ground running.
- c. **The eBird dataset:** Going through these resources will help you understand the ebird data structure and get you some knowledge of the car wheels, engine, and transmission workings, so that we can drive it.
 - i. State of India's Birds 2023: Viswanathan, Ashwin, Karthik Thrikkadeeri, Pradeep Koulgi, Praveen J., Arpit Deomurari, Ashish Jha, Ashwin Warudkar, et al. 2025. “State of India's Birds 2023: A Framework to Leverage Semi-Structured Citizen Science for Bird Conservation.” *Ecosphere* 16(7): e70290. <https://doi.org/10.1002/ecs2.70290>. **This paper provides an excellent example of how eBird data can be used at (very) large scales to infer population trends. It also is useful for understanding the data structure of eBird.**
 - ii. **To understand the biases of eBird data and a primer on how to counter some, refer to [Best practices for using eBird data](#):** Strimas-Mackey, M., W.M. Hochachka, V. Ruiz-Gutierrez, O.J. Robinson, E.T. Miller, T. Auer, S. Kelling, D. Fink, A. Johnston. 2023. *Best Practices for Using eBird Data*. Version 2.0. <https://ebird.github.io/ebird-best-practices/>. Cornell Lab of Ornithology, Ithaca, New York. <https://doi.org/10.5281/zenodo.3620739>.

3. Pre-workshop Exercises: We are now driving the car!

1. **A practice dataset: Please find the data for Bangalore for the year 2024 here:**
 - a. **The participants are expected to download this data from the provided link, and explore the dataset structure, distribution of sampling locations, and answer which area in bangalore is the most popular for birding?**
 - b. **Optional exercise: For a species of your choice in Bangalore create plots that demonstrate the seasonality of a species, or at least list down the steps required to do so.**
2. **Practice thinking:** the participants are expected to come up with an approach to answer the following questions and the drawbacks of using eBird data to answer such questions:
 - i. For a given region find areas which are promising for conservation but do not show up as important on the hotspot map.

- ii. See how birder behaviour and therefore the data is impacted by events such as COVID?
- iii. Create a graph of seasonality of Eastern Cattle Egret in Bangalore
- iv. For any species, how can one infer species trends from eBird data for 10 years?

Suggested Readings

Karthik Thrikkadeeri, Ashwin Viswanathan, Despite short-lived changes, COVID-19 pandemic had minimal large-scale impact on citizen science participation in India, *Ornithological Applications*, Volume 126, Issue 4, 4 November 2024, duae024, <https://doi.org/10.1093/ornithapp/duae024>

Menon, T., Ramesh, V., & Barve, S. (2023). Himalayan birds that show the greatest elevational shifts remain within the narrowest thermal regimes. *Global Ecology and Biogeography*, 32, 2111–2121. <https://doi.org/10.1111/geb.13761>

Girish, K. S., & Srinivasan, U. (2022). Community science data provide evidence for upward elevational range shifts by Eastern Himalayan birds. *Biotropica*, 54, 1457–1465. <https://doi.org/10.1111/btp.13133>

Viswanathan, Ashwin, Karthik Thrikkadeeri, Pradeep Koulgi, Praveen J., Arpit Deomurari, Ashish Jha, Ashwin Warudkar, et al. 2025. “State of India's Birds 2023: A Framework to Leverage Semi-Structured Citizen Science for Bird Conservation.” *Ecosphere* 16(7): e70290. <https://doi.org/10.1002/ecs2.70290>

Nameer, P. O. (2020). The expanding distribution of the Indian Peafowl (*Pavo cristatus*) as an indicator of changing climate in Kerala, southern India: A modelling study using MaxEnt. *Ecological Indicators*, 110, 105930.

Think about exercises 3.1 and 3.2, keeping in mind that the questions are open ended. We will choose one and go through the steps we would take to answer said question. We will share the code and you can follow along with us at the workshop.